Last time: for y a function of t was considering the differential equation:

$$ay'' + by' + cy = 0$$

And we were first considering the case when the characteristic equation (auxiliary equation), namely

$$ar^2 + br + c = 0$$

has two distinct real solutions, say r_1, r_2. Then we have general solution:

$$y(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}$$

So this a *second order* differential equation that seems to have two "distinct" solutions, namely $e^{r_1 t}$ and $e^{r_2 t}$

But what do we mean by "distinct"?

Say we had a tenth order linear differential equation. We would suspect that it has ten distinct solutions, but again what does that mean?

If say e^{t} and e^{2t} are both solutions, then perhaps $4e^{t} + 5e^{2t}$ is a solution as well (especially if the equation is homogeneous). But does that count as a "third" solution?

No! Part of the key with this is for the prior case

$e^{r_1 t}, e^{r_2 t}$ are <u>linearly independent.</u>

Familiar term of course but very different setting! A set of functions $f_1, f_2, \ldots, f_n$ is linearly independent if

$$c_1 f_1 + c_2 f_2 + \ldots + c_n f_n = 0$$

(note this means that function on left is identically 0, not just 0 at one point) forces

$$c_1 = c_2 = \ldots = c_n = 0$$

Again very similar looking definition to linear independence of vectors in $\mathbb{R}^n$ but unfortunately much harder to check here! But is a dilemma we need to consider.

If we are say solving an n-th order differential equation, we suspect it should have "n" solutions. But when we say n solutions we really mean n linearly independent solutions. But that means in theory we have to be able to check if a set of functions is such, however easy or hard it might be.

So if we have a set of n functions, denoted $f_1, f_2, \ldots, f_n$ how do we check if that set is linearly independent? We use the <u>Wronskian:</u>

$$\begin{vmatrix} f_1 & f_2 & \cdots & f_n \\ f_1' & f_2' & \cdots & f_n' \\ \vdots & \vdots & & \vdots \\ f_1^{(n-1)} & f_2^{(n-1)} & \cdots & f_n^{(n-1)} \end{vmatrix}$$

whether the determinant of this $n \times n$ matrix is identically 0 or not is key.

Ok in theory a lot to this. Why this? And what conditions must the functions satisfy? They must all be differentiable $n - 1$ times, but again a lot to this.

Not going to get heavily into this for several reasons, again quite involved if you try to examine it very carefully, but largely because in this class we mostly only solve second order equations so we don't really tend to need this in its general form.

Still good to note how it works and see that it too uses a linear algebra idea (determinant).

In second order, for y_1, y_2 we have this as:

$$\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

which becomes

$$y_1 y_2' - y_1' y_2$$

which is an important expression to keep in mind.

Turns out for y_1, y_2 solutions to same second order homogeneous differential equation (doesn't even have to have constant coefficients), y_1, y_2 are linearly dependent exactly when the Wronskian is zero at at least one point.

That's much weaker than saying it needs to be identically 0! Hence why useful, easier to check.

Why? Well without going heavily into it, if we differentiate the Wronskian, using the fact that y_1 , y_2 satisfy the same second order homogeneous equation will force the Wronskian's derivative to be a multiple of itself, making it an exponential style function which can only be 0 at a single point if it's identically 0.

Back to solving:

$$ay'' + by' + cy = 0$$

Characteristic equation is

$$ar^2 + br + c = 0$$

(1) When this has two distinct real solutions r_1 , r_2 general solution is:

$$y(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}$$

Next case is

(2) Say characteristic equation has repeated real solution r. Then general solution is

$$y(t) = c_1 e^{rt} + c_2 te^{rt}$$

That is, e^{rt} , te^{rt} are the two linearly independent solutions.

You likely learned that trick of multiplying by t when repeated solution in calculus. But why is that the idea?

Sure you can check it works, but what is the fundamental idea? Turns out that too involves linear algebra ideas - something we might clarify a bit more later on.

B4.2.3 $$y'' + 5y' + 6y = 0$$

Characteristic equation:

$$r^2 + 5r + 6 = 0$$

----> $$(r + 2)(r + 3) = 0$$

----> $$r = -3, -2$$

Since roots are distinct, general solution is:

$$y(t) = c_1 e^{-3t} + c_2 e^{-2t}$$

B4.2.9 $4y'' - 4y' + y = 0$

Characteristic equation:

$$4r^2 - 4r + 1 = 0$$

----> $$(2r - 1)^2 = 0$$

----> $$r = \frac{1}{2}, \frac{1}{2}$$

Since repeated real root here, general solution becomes

$$y(t) = c_1 e^{\frac{1}{2}t} + c_2 te^{\frac{1}{2}t}$$

B4.3 Nonreal Roots to Charcateristic Equation

More precisely second order linear homogeneous equations that have constant coefficients for which the characteristic equation has nonreal roots.

That is, when examining $ay'' + by' + cy = 0$, say the characteristic equation $ar^2 + br + c = 0$ has solutions $r = \alpha \pm \beta i$ for some $\beta \neq 0$

What is general solution? Last section suggests

$$y(t) = Ce^{(\alpha+\beta i)t} + De^{(\alpha-\beta i)t}$$

But realistically we can't leave it in this form, as the differential equation is real so solution ought to be a real function. And if we wanted to leave it here, how would we check it's correct say? Trying to differentiate it with the i mixed in, iffy at best.

Plus given the i in the exponent we probably need to allow C and D to have i in them as well which seems like it will complicate things further.

But turns out C and D being such is actually in a sense what allows us to fix this. But how?

Take $C = a_1 + b_1 i$, $D = a_2 + b_2 i$ and use important identity (Euler's formula):

$$e^{it} = \cos t + i \sin t$$

which is proved using Maclaurin Series for e^x, $\cos x$, $\sin x$. Gives the formula

$$e^{i\pi} = -1$$

which many call the most remarkable formula in math. After all the three probably most used special symbols coming together in such a simple looking formula!

But key here is that enables us to bring the i down from the power to the front which is easier to handle.

So doing that gives here

$$(a_1 + b_1 i)e^{\alpha t}\,[\cos(\beta t) + i\sin(\beta t)]$$
$$-$$
$$(a_2 + b_2 i)e^{\alpha t}\,[\cos(\beta t) - i\sin(\beta t)]$$

Now making the imaginary terms 0 we must have

$$(b_1 + b_2)e^{\alpha t}\cos(\beta t) = 0 \quad ,$$

$$(a_1 - a_2)e^{\alpha t}\sin(\beta t) = 0$$

so that we must have $a_1 = a_2$, $b_1 = -b_2$ for the function to be real. Then overall it becomes:

$$e^{\alpha t}\,[2a_1\cos(\beta t) - 2b_1\sin(\beta t)]$$

so for simplicity, we write the general solution as

$$y(t) = c_1 e^{\alpha t} \cos(\beta t) + c_2 e^{\alpha t} \sin(\beta t)$$

where c_1, c_2 are any real numbers.

So now trig functions involved when they weren't when the solution(s) were real. Interestingly, kind of similar to when we were trying to diagonalize 2 x 2 real matrices, when we had nonreal eigenvalues the "pseudo-diagonalization" we used instead (Section 5.5) also involved using trig functions to help.

So we see why allowing C and D to be nonreal was important. If we hadn't, then we would have needed $C = D$ to make sure the imaginary part cancels, so we would have only had one unknown constant / free variable. Not natural a second order equation, we feel there ought to be two of them.

By allowing them to be nonreal, that gave us four free variables to start and then forced two conditions, making the coefficients of both $e^{\alpha t}\cos(\beta t)$ and $e^{\alpha t}\sin(\beta t)$ equal to 0.

So we then wind up with two free variables on the whole, as it should be.

B4.3.1

$$y'' + 9y = 0$$

Characteristic equation:

$$r^2 + 9 = 0$$

---> $$r = \pm 3i$$

so that $y(t) = c_1 \cos(3t) + c_2 \sin(3t)$

is the general solution.

B4.3.3

$$z'' - 6z' + 10z = 0$$

Characteristic equation:

$$r^2 - 6r + 10 = 0$$

----> $$(r - 3)^2 + 1 = 0$$

----> $$r = 3 \pm i$$

Hence $y(t) = c_1 e^{3t} \cos t + c_2 e^{3t} \sin t$

is the general solution.

Note how different the solutions to these problems are as one has exponential term, other does not. Makes the graphs of the two very different! Quite dissimilar physical situations these are.

Likely saw this in Calculus, motion of a spring say, issues of if being simple harmonic motion versus there being a damping force.

B4.4 Nonhomogeneous (with Constant Coefficients)

$$ay'' + by' + cy = f(t)$$

How to find general solution to this? Main approach:

(1) Find general solution to homogeneous counterpart, i.e., $ay'' + by' + cy = 0$, sometimes called the *complementary equation* which is what we did in last two sections, that solution often denoted as y_c (or y_h).

(2) Then find one particular solution to the nonhomogeneous equation, denoted y_p. Then overall general solution is

$$y = y_c + y_p$$

Really same thing we did with matrix equations $Ax = 0$ versus $Ax = b$. So main thing here is how to find the particular solution.

Doing this for this style of second order differential equation can unfortunately be very hard, depends a lot on $f(t)$. Not like $Ax = b$ which basically used same strategy no matter what b was.

Will try to do it for a few different $f(t)$ to get started

(1) If $f(t)$ is a polynomial, usually try y_p to be a polynomial of the same degree.

(2) For k constant, if $f(t) = e^{kt}$, usually try $y_p = Ae^{kt}$

(3) If $f(t) = \cos(kt)$ or $\sin(kt)$, usually try $y_p = A\cos(kt) + B\sin(kt)$

Note in third do need both terms due to way cos, sin alternate between each other in derivatives.

While these may not always work exactly as suggested, they are all logical in that you will have same style of terms on both sides so logical that you have a chance to make them equal. It wouldn't really be logical if $f(t)$ were a polynomial to try $y_p = Ae^{kt}$, as the two sides would then not have the same style terms at all, wouldn't be logical that they could be equal.

For sum/difference of such terms in $f(t)$, luckily can just add the prospective y_p's since derivative of sum is sum of derivatives.

Even with product still can multiply prospective solutions since product rule behaves well enough with these style terms to allow it (they stay similar form when they are differentiated).

With quotient though that doesn't tend to work as in that rule you square the denominator, and squaring can be different enough to cause this to not be viable.

Also said above that we "usually" try such and such. That language suggests that it might not work in some situations. But which ones and why?

In short, there is a lot of nuance in this that we have to wade through it carefully.

Will expand more on this next time.